

DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B

Total Parameters

21B

Activated per Token

128K

Context Length

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

MHA

KV Cache Bottleneck

Standard MHA stores full K/V pairs per head per token. At 128K context, cache size becomes prohibitive for autoregressive generation.

Dense

Training Cost Crisis

Dense models require activating all parameters. A 70B dense model costs 3-5× more to train than a comparable MoE architecture.

GQA

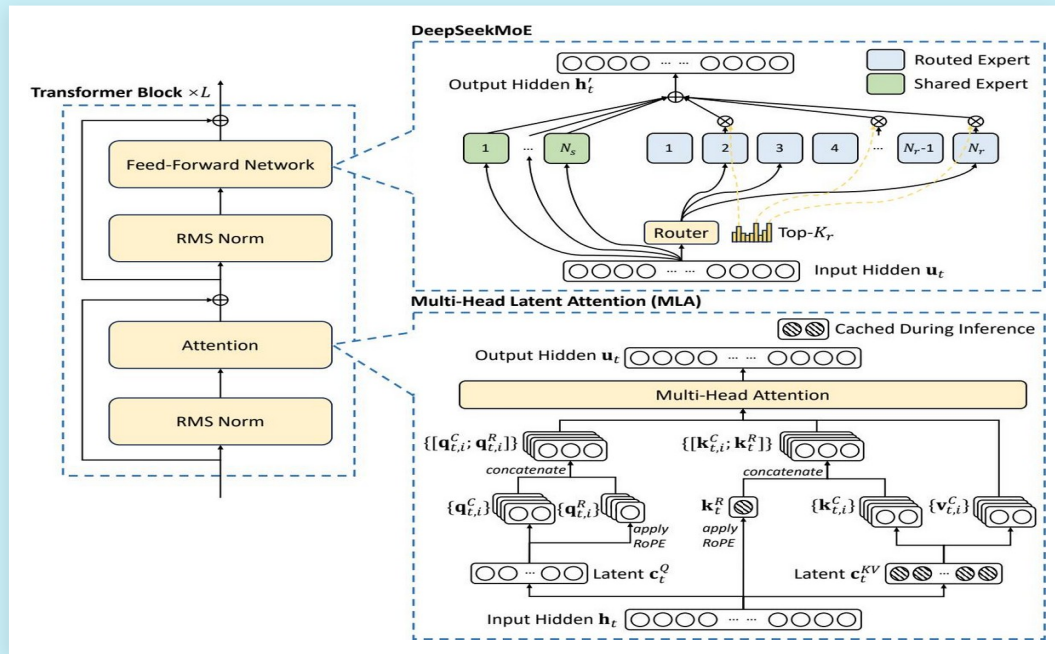
Efficiency Trade-offs

GQA and MQA reduce cache at the cost of model quality — they are approximations, not solutions.

Need: Architecture innovations that reduce costs without sacrificing performance

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block



Multi-Head Latent Attention

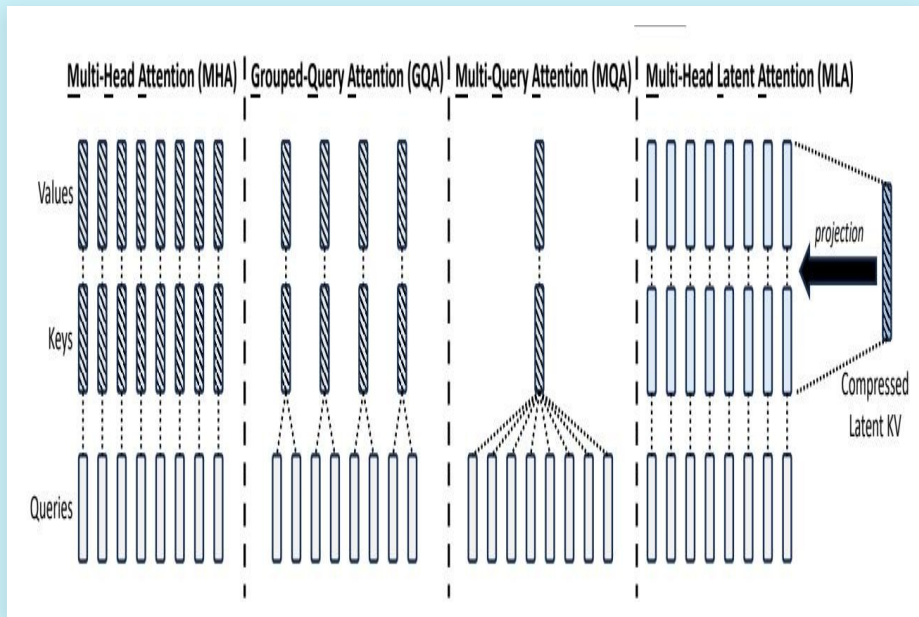
- Low-rank KV compression
- Decoupled RoPE positional key
- On-the-fly K/V reconstruction

DeepSeekMoE FFN

- Shared experts — always active
- Routed experts — top-K selection
- Fine-grained expert segmentation
- Device-limited routing

Multi-Head Latent Attention

Low-Rank Key-Value Joint Compression — 93.3% KV Cache Reduction



Problem: MHA stores full KV per head per token — cache scales with context length $\times$ num_heads $\times$ dim

Compress

K and V are jointly projected into a low-rank latent vector c_t^{KV} and cached

Reconstruct

Full K and V are recovered on-the-fly via learned projection matrices at each layer

RoPE

Decoupled position key with RoPE enables positional awareness without

Result: Superior performance to MHA while requiring dramatically less cache

DeepSeekMoE

Fine-Grained Experts with Shared Expert Isolation — 42.5% Training Cost Reduction

Shared Experts (N_s)

- Always activated — capture common knowledge
- Ensure minimum expertise coverage across all tokens
- Reduce redundancy in routed expert selection

Routed Experts (N_r)

- Sparsely activated via top- K_r router
- Fine-grained segmentation increases specialization
- Only a fraction activated per token — massive compute savings

Device-Limited Routing

Each token routed to experts on at most M devices, minimizing cross-node communication overhead.

Auxiliary Load-Balance Loss

Prevents expert collapse by penalizing uneven token distribution across experts and devices.

Token-Dropping Strategy

Dynamically drops tokens during training to prevent expert overflow — does not affect inference.

Fine-Grained Expert Segmentation

Breaking experts into smaller units increases specialization potential and routing flexibility.

Efficiency Gains vs. DeepSeek 67B

DeepSeek-V2 achieves dramatic efficiency improvements over its dense predecessor

42.5%

Training Cost Reduction

vs. DeepSeek 67B

DeepSeekMoE architecture enables training a 236B model at a fraction of dense model cost

93.3%

KV Cache Reduction

vs. MHA baseline

MLA low-rank compression dramatically reduces memory footprint during generation

5.76x

Generation Throughput

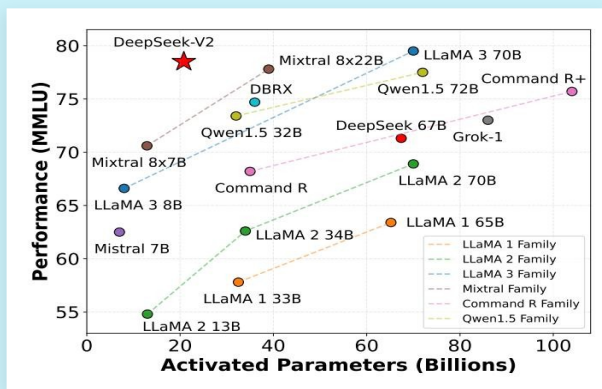
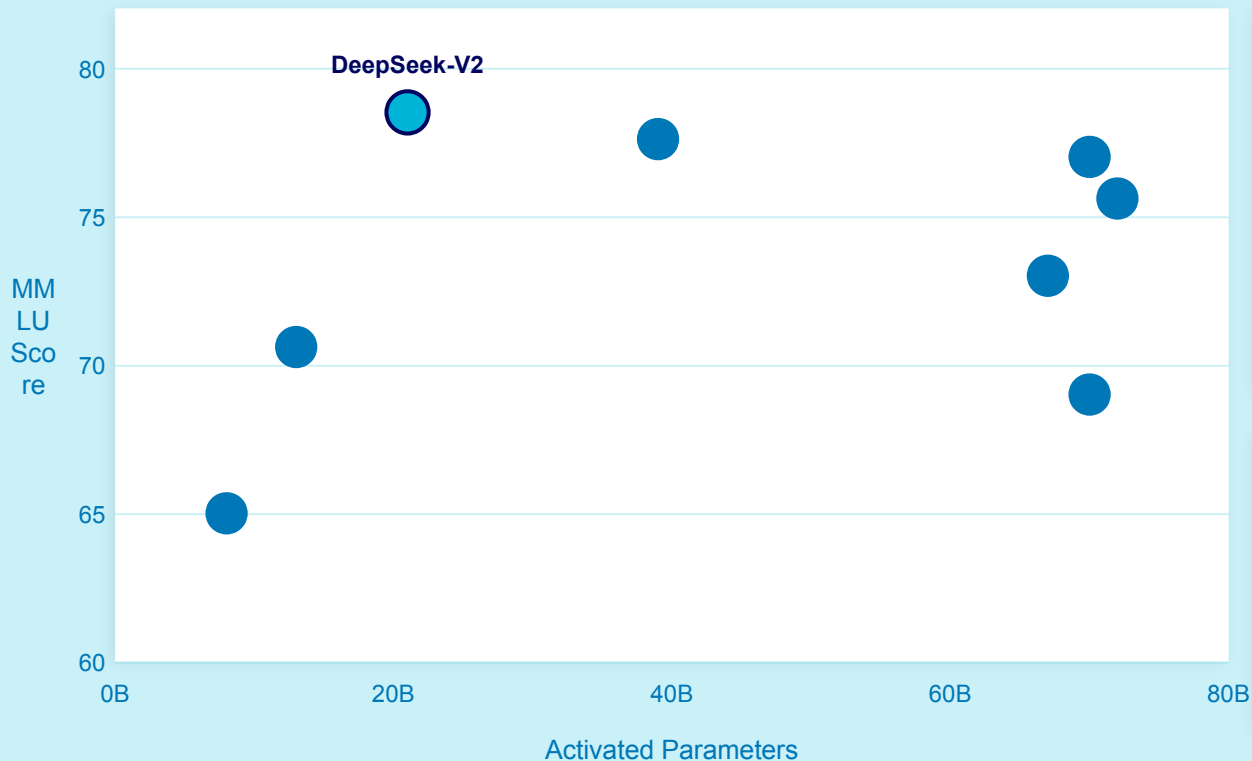
max generation throughput

Reduced cache size and sparse activation enable much higher throughput in practice

DeepSeek-V2: 236B total / 21B activated vs. DeepSeek 67B: 67B dense — same quality tier, different efficiency class

MMLU Performance vs. Activated Parameters

DeepSeek-V2 achieves top-tier MMLU with far fewer activated parameters — Pareto-optimal efficiency



Source: arxiv.org/abs/2405.04434

Key Comparison Data

DeepSeek-V2	21B	78.5
LLaMA 3 70B	70B	77.0
Mixtral 8x22B	39B	77.6
Qwen1.5 72B	72B	75.6

DeepSeek-V2 Chat (RL) Alignment Results

SFT on 1.5M conversational sessions + GRPO reinforcement learning

AlpacaEval 2.0

Length-Controlled Win Rate

38.9%

Outperforms most open-source models;
competitive with closed-source systems

MT-Bench

Overall Score

8.97

Strong multi-turn instruction following across
diverse task categories

AlignBench

Overall (Chinese)

7.91

Outperforms all open-source and most closed-source models in Chinese



Strongest open-source chat model in Chinese — outperforms all open-source and most closed-source models

Post-training: Supervised Fine-Tuning (1.5M sessions) → GRPO-based Reinforcement Learning

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

01

MLA — Superior KV-Cache Solution

Achieves better performance than MHA while reducing KV cache by 93.3%. A strictly better alternative to GQA/MQA.

02

DeepSeekMoE — Economical Training

Fine-grained experts with shared expert isolation enable training a 236B model at 42.5% lower cost than a 67B dense model.

03

Top-Tier Quality at Fraction of Activated Params

Only 21B activated parameters match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.

04

Strong Multilingual Alignment

DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English.

05

Open-Source Availability

Pareto-optimal positioning: best performance-per-parameter in its class, enabling efficient large-scale deployment.